VASTRAVIEW: DETAILED SYSTEM ARCHITECTURE

Full-Stack Multilayer Topology: Edge CPU Anthropometry & Remote Cloud GPU Diffusion VTON

1. CLIENT PRESENTATION TIER (React 19.1 + TypeScript 5.8 + Vite 6.4 — Port 5173)

Acquisition & Ingestion

- HTML5 Webcam Live Stream Capture
- Drag-and-Drop Image Uploader
- EXIF Orientation & Mime Audit
- Client Bounds: 256px - 12MB limit

Skeletal & Silhouette Canvas

- 33 3D Landmark Vector Visualization
- Interactive Joint Adjustment Overlay
- Real-Time GrabCut Contour Lines
- Camera Silhouette Guides

Brand Sizing Workspace

- Dynamic Multi-Brand Selector
- Fit Preference (Tight/Regular/Loose)
- Fit Score (%) & Dimension Indicators
- Dimension Breakdown (Chest, Waist)

Comparison Try-On Studio

- Interactive Split Slider (Before/After)
- Dual Pan & Zoom Coordinate Sync
- Curated 57-Garment Studio Wardrobe
- Lossless PNG / WebP Export

REST JSON / Multipart | **33-Keypoints & Bounds** | **Lossless PNG Render**

2. CORE APPLICATION & INTELLIGENCE TIER (Python 3.12 + FastAPI 0.162 + Uvicorn — Port 8000)

API Gateway & Security

- Argon2id Password Hashing
- SameSite HttpOnly Auth Cookies
- In-Memory Short-Lived JWT
- Strict Pydantic v2 Request Filters
- Health Probes (/api/v1/health)

Anthropometry Engine (CPU)

- MediaPipe Pose Landmarker Full
- 33 Keypoints (x,y,z) with v > 0.65
- Stature Ground Scale: s = H_true/H_px
- Ramanujan Elliptical Slice Model:
  Circumf = π[3(a+b) - √((3a+b)(a+3b))]
- Slices: Shoulder, Chest, Waist, Hip

Deterministic Sizing Engine

- Midpoint Relative Euclidean Metric:
  Error_s = Σ w_m * |d_m - midpoint| / tol
- Weight Vector W:
  Tops: Chest 0.50, Shld 0.35, Wst 0.15
  Bottoms: Waist 0.50, Hip 0.35, Inseam 0.15
- Fit Indicators (Ideal, Snug, Loose)

VTON Orchestrator

- Asynchronous Queue (.runtime/)
- Lossless WebP Payload Packing
- Category Garment Parse Masking
- Lossless Identity Compositor:
  Re-pastes Face, Skin & Background
- Self-Healing Tunnel Monitor

PyMongo / Relational Queries | **Encrypted Tunnel / Job Dispatch**

3. LOCAL PERSISTENCE & ASSET TIER

MongoDB Community 8.2 (Port 27018 / PyMongo)

- 23 Sourced Brand Size Charts
  (sourced-body-charts-v2.json)
- Atomic (brand, category) indexes
- User Biometric Scan Profiles
- Private User Wardrobe Collections

PostgreSQL 17 & Filesystem (Port 5433 / SQLAlchemy)

- Asynchronous Try-On Job States
- Pipeline Execution Audit Logs
- File Store: /data/processed/
- Curated 57 Reference Garments
- Model Weights: pose_landmarker.task

4. REMOTE GPU CLUSTER (Google Colab T4 / A100)

Reverse Tunnel Gateway (Cloudflare / ngrok)

- Zero Cloud Storage Requirement
- Self-Healing Reconnect Bridge
- Bearer Token Authentication
- 3-Second Heartbeat Latency Poll

FASHN VTON 1.5 Worker (PyTorch / FP16 CUDA)

- online_worker.py (Port 8080)
- 576x864 Latent Diffusion Inpainting
- Deep Garment Deformation Network
- 8.4s Inference per Generation
- Lossless WebP Output Encoding